Harbor Point Solar Cooperative

Operations Handbook 2025

This handbook describes how the Harbor Point Solar Cooperative runs its generation site, maintains its equipment, responds to faults and outages, and reports to its members. It is written for field technicians, control room staff and members of the cooperative board.

1. About the Cooperative

Harbor Point Solar Cooperative was founded in 2017 by 212 households that wanted local, member-owned power. Each member holds at least one share, and shares are priced at $250. The cooperative is governed by a seven-person board elected every two years, and every member has one vote regardless of how many shares they hold.

In 2025 the board approved a dividend of $41.50 per member share, paid in January 2026. Surplus income beyond the dividend is placed in an equipment reserve fund used for inverter replacement and battery upgrades.

2. Generation Site and Equipment

The main generation site is the Kestrel Ridge Array, which has a rated capacity of 14.6 MW and consists of 38,400 panels arranged in 96 rows. Power from the panels is collected in combiner boxes and converted to grid-compatible AC power by 48 string inverters housed along the eastern fence line.

Energy storage is provided by the Tern battery system, a lithium iron phosphate installation with a capacity of 9.2 MWh. The Tern system was commissioned in March 2023 and is used to cover evening demand and to supply priority loads during outages.

During 2025 the array generated 24,180 MWh of electricity. This corresponds to a capacity factor of 18.9 percent, which is within the range expected for a fixed-tilt array at this latitude.

3. Routine Maintenance

Inverters are inspected every 97 days. Each inspection covers cooling fans, filter condition, terminal torque and firmware version. Technicians record results in the maintenance log before leaving the inverter housing.

Panels are cleaned twice a year, in April and in October. Cleaning uses deionized water at roughly 0.8 liters per panel, applied with soft brushes. Detergents are never used because residue attracts dust and reduces output. Cleaning is scheduled for early morning when panels are cool, since cold water on hot glass can cause thermal stress.

Vegetation along the perimeter is cut back to keep a clear margin of three meters from the fence, which reduces both shading and fire risk during the dry season.

4. Fault Codes

The monitoring system reports faults using numeric codes. Technicians should acknowledge every fault in the control system before starting any physical work.

Code E-17 indicates inverter overheating. An inverter reporting E-17 shuts down automatically when its internal temperature reaches 78 degrees Celsius, and it restarts only after cooling below 65 degrees Celsius and being manually reset by a technician.

Code E-42 indicates a grid frequency deviation. It is raised when the grid frequency moves more than 0.5 Hz away from nominal, and the affected inverters disconnect from the grid until the frequency is stable again.

Code E-63 indicates battery cell imbalance. It is raised when the charge difference between the highest and lowest cells in a Tern battery module exceeds 4 percent. The module is isolated and a balancing cycle is started.

5. Safety Procedures

All electrical work follows lockout-tagout procedure. A technician must apply a personal lock and tag to the isolation point and verify absence of voltage before touching any conductor.

Combiner boxes have an arc-flash boundary of 1.2 meters. Anyone working inside this boundary must wear arc-rated clothing and a face shield. Work on energized equipment requires written approval from the site supervisor.

Lightning is treated as an emergency. When a strike is detected within 8 kilometers, all outdoor work stops and staff move into the control building until 30 minutes have passed since the last strike.

6. Outage Response

When the grid fails, the cooperative isolates the site and runs a priority restoration plan. Tier 1 loads are the Harbor Point Clinic and Water Pumping Station No. 3. Power to Tier 1 loads must be restored within 45 minutes, using the Tern battery if the array cannot supply them directly.

Tier 2 loads are the elementary school and the community shelter, restored within 3 hours. Remaining member households are restored after Tier 1 and Tier 2 are stable. Members are notified by text message at the start of an outage and every two hours afterward.

The control room can be reached on extension 4410 and is staffed around the clock.

7. Reporting to Members

The cooperative publishes an annual report each February. The report includes total generation, capacity factor, outage minutes, the dividend decision and the equipment reserve balance. Members may request the full maintenance log by written request to the board secretary, and the log is provided within 14 days.